

Xinyu Xiao

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Education

Fudan University, Shanghai, China

September 2026 — June 2031 (Expected)

Ph.D. in Computer Science and Technology

Advisor: Prof. Tao Xie

Huazhong University of Science and Technology, Wuhan, China

September 2022 — June 2026

B.Eng. in Cyberspace Security

Cumulative GPA: 4.31/5.00, Overall Rank: 15/105

Project Experience

Cross-Abstraction-Level Agentic Operator Programming System

Shanghai, China

Core Developer

June 2026 – Present

Designed and developed an LLM-driven automatic operator optimization system for AMD GPUs and Ascend NPUs, enabling cross-level development across TileLang, HIP C++, and AscendC. Implemented performance diagnosis, dynamic abstraction-level selection, and candidate rollback mechanisms, allowing agents to autonomously choose between high-level structural modifications and low-level code optimizations based on runtime feedback. The related paper is currently under review.

Design and Implementation of a Deep Neural Network Inference System Based on High-Performance TileLang Operators

Wuhan, China

Undergraduate Thesis, Huazhong University of Science and Technology; Advisor: Associate Researcher Yunhe Zhang

December 2025 – May 2026

To address GPU memory access bottlenecks caused by concurrent LLM invocations in multi-agent systems, developed a highly optimized inference operator library based on TileLang for Qwen3-32B and built a high-throughput end-to-end prototype system supporting concurrent inference for multiple agents.

Microsecond-Level GPU Preemption System

Beijing, China

Research Intern, Institute of Computing Technology, Chinese Academy of Sciences; Advisor: Associate Researcher Chenxi Wang

March 2024 – July 2025

Participated in the development of a service-level-objective (SLO)-oriented GPU preemption system for microsecond-scale scheduling. The paper has been submitted to **ACM TACO**.

Internship Experience

TileLang

Beijing, China

Research Intern, Peking University; Advisor: Associate Researcher Zhi Yang

March 2025 – August 2025

Developed and optimized language features and operators for TileLang.

AMD

Shanghai, China

AI/ML Optimization Intern

July 2025 – December 2025

Worked on the AMD ROCm framework and software stack and contributed to the development of its operator ecosystem.

Awards and Honors

Science and Technology Innovation Scholarship

October 2024

Huazhong University of Science and Technology

Academic Excellence Scholarship

October 2023

Huazhong University of Science and Technology

Gold Award (1st Place)

June 2025

Hubei Information Technology Application Innovation Competition

First Prize

November 2024

National College Students Mathematical Modeling Contest, Hubei Division

Second Prize

May 2024

National English Competition for College Students

Third Prize

August 2024

“China Software Cup” University Software Design Competition

Selected Coursework

Course	Grade
• Introduction to Artificial Intelligence	• 98 5.0
• Principles of Database Systems	• 93 4.8
• Principles of Operating Systems	• 92 4.7
• Principles of Cryptography	• 92 4.7
• Computer Communications and Networks	• 89 4.4
• Design and Analysis of Algorithms	• 88 4.3

Skills

- **Domain Knowledge:** Operating Systems and GPU Systems, Agent Systems Development, AI Infrastructure and Applications, Cryptography and Information Security
- **English:** CET-4: 676/710 (Top 1%), CET-6: 607/710 (Top 10%)
- **Programming Languages:** C++, Python, CUDA, Go, Rust, Assembly